

## 6 Package information

To meet environmental requirements, ST offers these devices in different grades of **ECOPACK** packages, depending on their level of environmental compliance. ECOPACK specifications, grade definitions and product status are available at: [www.st.com](http://www.st.com). ECOPACK is an ST trademark.

### 6.1 Device marking

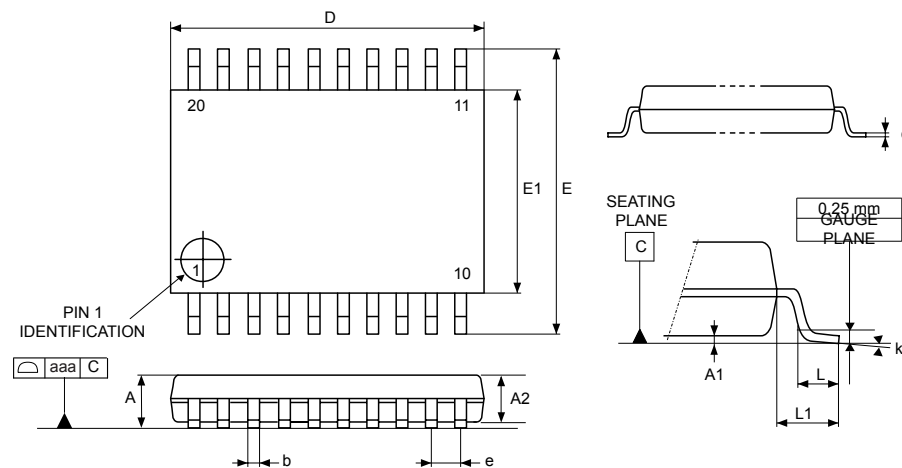
Refer to technical note "Reference device marking schematics for STM32 microcontrollers and microprocessors" (TN1433) available on [www.st.com](http://www.st.com), for the location of pin 1 / ball A1 as well as the location and orientation of the marking areas versus pin 1 / ball A1.

Parts marked as "ES", "E" or accompanied by an engineering sample notification letter, are not yet qualified and therefore not approved for use in production. ST is not responsible for any consequences resulting from such use. In no event will ST be liable for the customer using any of these engineering samples in production. ST's Quality department must be contacted prior to any decision to use these engineering samples to run a qualification activity.

A WLCSP simplified marking example (if any) is provided in the corresponding package information subsection.

### 6.2 TSSOP20 package information (YA)

**Figure 37. TSSOP20 - Outline**



YA\_ME\_V3

**Table 71. TSSOP20 - Mechanical data**

| Symbol | Millimeters |      |      | Inches <sup>(1)</sup> |        |        |
|--------|-------------|------|------|-----------------------|--------|--------|
|        | Min         | Typ  | Max  | Min                   | Typ    | Max    |
| A      | -           | -    | 1.2  | -                     | -      | 0.0472 |
| A1     | 0.05        | -    | 0.15 | 0.002                 | -      | 0.0059 |
| A2     | 0.8         | 1    | 1.05 | 0.0315                | 0.0394 | 0.0413 |
| b      | 0.19        | -    | 0.3  | 0.0075                | -      | 0.0118 |
| c      | 0.09        | -    | 0.2  | 0.0035                | -      | 0.0079 |
| D      | 6.4         | 6.5  | 6.6  | 0.252                 | 0.2559 | 0.2598 |
| E      | 6.2         | 6.4  | 6.6  | 0.2441                | 0.252  | 0.2598 |
| E1     | 4.3         | 4.4  | 4.5  | 0.1693                | 0.1732 | 0.1772 |
| e      | -           | 0.65 | -    | -                     | 0.0256 | -      |
| L      | 0.45        | 0.6  | 0.75 | 0.0177                | 0.0236 | 0.0295 |